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Experimental investigation and thermodynamic re-assessment of the Mn–Gd binary system

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Abstract In this work, 13 Mn–Gd alloys prepared by arc-melting method were investigated experimentally by means of differential thermal analysis. The temperatures of the invariant reactions and liquidus in the Mn–Gd binary system were determined according to the measured thermal analysis curves. One key as-cast $Mn_{70}Gd_{30}$ alloy was examined using differential thermal analysis and scanning electron microscopy equipped with energy-dispersive X-ray spectroscopy to confirm the type of the invariant reaction ($L + Mn_{23}Gd_6 \leftrightarrow Mn_2Gd$). Based on the critical review of the available experimental data measured in the present work and in the reported literature, the Mn–Gd binary system was assessed thermodynamically using the CALPHAD method. The solid solution phases including liquid, α-Mn, β-Mn, γ-Mn, δ-Mn, α-Gd and β-Gd are modeled by the substitutional solution model, while the intermetallic compounds such as $GdMn_2$, Gd_6Mn_{23} and $GdMn_{12}$ are treated as the stoichiometric compounds. A set of self-consistent thermodynamic parameters obtained finally to describe the Gibbs energies of various phases in the Mn–Gd binary system can be used to reproduce well phase equilibria and thermodynamic data.

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Introduction

Ternary intermetallic compounds containing rare earth (RE) elements, transition metals (e.g., Mn, Cu, Co, Fe) and third elements (e.g., Si, Ge, Sn) have been investigated extensively because of their interesting magnetic properties [1–3]. These stable ternary intermetallic compounds with particular crystal structures in the RE–Mn–X (X = Si, Ge, Sn, etc.) systems show complex magnetic properties due to the interesting interplay between the 3*d* and 4*f* magnetism and the strong dependence of the magnitude of the Mn moment and the magnetic state of the Mn sublattice on the Mn–Mn interatomic distances [4, 5]. For example, ternary REMnSi (RE = La–Sm, Gd) compounds are CeFeSi-typed structure or Co_2Si-typed structure (RE = Tb–Er), while ternary REMnGe (R = Gd–Tm) compounds are TiNiSi-typed structure [6–8]. On the other hand, ternary REMnX compounds with CeFeSi-typed structure (e.g., GdMnSi) show the high change of the magnetic entropy and spin glass behavior [9, 10]. Ternary $REMn_2X_2$ compounds with the body-centered tetragonal $ThCr_2Si$-typed structure (e.g., $GdMn_2Si_2$, $GdMn_2Ge_2$ and $NdMn_2Si_2$) exhibit magnetocaloric effect [11, 12]. Ternary $REMn_6X_6$ compounds with $HfFe_6Ge_6$-type structure (e.g., $GdMn_6Sn_6$, $GdMn_6Ge_6$ and $ErMn_6Sn_6$) show thermal expansion, spontaneous magnetostriction and magnetoresistance [13–16].

To explore better ternary intermetallic compounds with desirable magnetic properties in the RE–Mn–X (X = Si, Ge, Sn, etc.) ternary systems, the knowledge of phase equilibria and thermodynamic properties of the related RE–Mn-based alloy system is indispensable. As

key sub-binary systems of the RE–Mn–X (X = Si, Ge, Sn, etc.) ternary systems, the Mn–RE binary systems [17] and RE–X binary systems (e.g., Ho–Sn [18], Yb–Sn [19]) were assessed thermodynamically. In particular, based on the critical review of the reported experimental data and the measured phase diagram data, the Mn–Nd binary system was re-assessed thermodynamically in our previous work [20]. The Mn–Gd binary system was optimized by Gröbner et al. [21] only using the experimental data measured by Kirchmayr and Lugscheider [22]. The enthalpies of mixing of liquid Mn–Gd alloys at 1650 K measured recently by Ivaov et al. [23] and the heat capacity of Mn_2Gd determined by Marcos et al. [24] were not used in their optimization. Using the modified quasichemical model to describe the liquid phase and new experimental data obtained by Antion [25], Kim and Jung [26] re-optimized the Mn–Gd binary system. In these two assessments mentioned above, there is a contradiction that Kim and Jung [26] took the congruent melting of Mn_2Gd into consideration, while Gröbner et al. [21] suggested the formation of Mn_2Gd through a peritectic reaction at 1214 K. Further experimental investigation and thermodynamic re-assessment of the Mn–Gd binary system are still necessary to develop compatible thermodynamic database of the Mn–RE multi-component systems.

Therefore, in this work, the Mn–Gd binary system was investigated experimentally and then was re-assessed thermodynamically using the CALPHAD method [27] based on critical review of the experimental phase equilibria data and thermodynamic data.

Literature information

Phase diagram data

The phase diagram of the Mn–Gd binary system was studied firstly by Kirchmayr and Lugscheider [22] by means of differential thermal analysis (DTA) and X-ray diffraction (XRD). Three stable intermetallic phases, $Mn_{12}Gd$, $Mn_{23}Gd_6$ and Mn_2Gd, were found. According to the review of the Mn–Gd binary system by Okamoto [28], it was indicated that the liquidus of the rich-Gd part is not reasonable according to the fact that no solubility or only very limited solubility of Mn in rare earth elements was reported. Recently, the Mn–Gd binary system was studied by Antion [25] using differential thermal analysis (DTA) and metallography. The peritectic reaction, $L + Mn_{23}Gd_6 \leftrightarrow Mn_2Gd$, was not found by Antion [25], which is not consistent with the experimental results [22]. In addition, the eutectic temperature of the invariant reaction, $L \leftrightarrow Mn_2Gd + \alpha\text{-Gd}$, was determined to be 1085 K by Antion [25], while it was measured to be 1103 K by Kirchmayr and Lugscheider [22].

Thermodynamic data

Nikolaenko and Nosova [29] measured the enthalpies of mixing of liquid Mn–Gd alloys and the partial enthalpies of mixing of Gd ($0 < x_{Gd} < 0.27$) and Mn ($0.45 < x_{Gd} < 1.0$) in liquid alloys at 1600 K by heat flux high isoperibolic calorimetry. Ivanov et al. [23] employed the isoperibolic calorimetry method to measure the enthalpies of mixing of the liquid Mn–Gd alloys and the partial enthalpies of mixing of Gd ($0 < x_{Gd} < 0.38$) and Mn ($0.45 < x_{Gd} < 1.0$) at 1650 K. The enthalpies of mixing of the Mn–Gd liquid alloys show negative deviation in the whole composition range. The maximum values of the enthalpy of mixing of the Mn–Gd liquid alloys are about -3.62 kJ mol^{-1}. However, the experimental data measured by Nikolaenko and Nosova [29] show large position deviation. The experimental data determined by Ivanov et al. [23] are also consistent with the enthalpies of mixing of liquid Mn–RE alloys [30]. Consequently, the experimental data determined by Ivanov et al. [23] were taken into account in the present work.

Up to now, the enthalpies of formation of the intermetallic compounds (Mn_2Gd, $Mn_{23}Gd_6$ and $Mn_{12}Gd$) in the Mn–Gd binary system were not determined experimentally in the literature. Marcos et al. [24] measured the heat capacity of Mn_2Gd intermetallic compound at low temperature range (1.8–300 K) using the Quantum Design Physical Property Measurement System (PPMS). The magnetic transformation temperature of Mn_2Gd was reported to be 105 K [24].

Experimental

Mn and Gd (99.99% purity, China New Metal Materials Technology Company, Ltd.) were used as starting materials. The alloys with the mass about 3 g were prepared by arc-melting pure elements under argon atmosphere using a nonconsumable tungsten electrode and were re-melted several times in order to improve their homogeneity. To achieve compositional homogeneity, the alloys (except for $Mn_{70}Gd_{30}$ alloy) were annealed in an evacuated quartz tube at 873 K for 336 h and then were quenched in ice water.

The microstructure examination and composition analysis of alloy samples (about 1 g) were measured by scanning electron microscopy with energy-dispersive X-ray analysis (SEM/EDS, FEI 450G). Thermal analysis measurements of the alloy samples (about 15 mg) were investigated in Al_2O_3 crucibles under a flowing pure N_2 atmosphere using differential thermal analysis (DTA, TA

Instruments SDT/Q-600) at different heating and cooling rates of 10 and 20 K min^{-1}, respectively.

Thermodynamic models

Pure elements

In the Mn–Gd binary system, the Gibbs energy for the element i in ϕ status is given as:

$$\begin{aligned} {}^{0}G_{i}^{\phi}(T) &= G_{i}^{\phi}(T) - H_{i}^{SER} \\ &= a + bT + cT\ln T + dT^{2} + eT^{3} + fT^{-1} + gT^{7} \\ &\quad + hT^{-9} \end{aligned} \tag{1}$$

in which $G_{i}^{\phi}(T)$ stands for the Gibbs energy of the element i (i = Gd, Mn) with the structure ϕ. H_{i}^{SER} is the enthalpy of the element i in standard reference state (298.15 K, 1 bar). T is the absolute temperature in Kelvin. ${}^{0}G_{i}^{\phi}(T)$ is the molar Gibbs energy of element i in the structure of ϕ referred to the enthalpy of its stable state at 298.15 K and 1 bar. In this work, the Gibbs energies of the elements, ${}^{0}G_{Gd}^{\phi}(T)$ and ${}^{0}G_{Mn}^{\phi}(T)$, are taken from the Scientific Group Thermodata Europe (SGTE) compiled by Dinsdale [30].

Solution phases

The solution phase ϕ (ϕ = liquid, α-Mn(cbcc-A12), β-Mn(cub-A13), γ-Mn(fcc-A1), δ-Mn(bcc-A2), α-Gd(hcp-A3) and β-Gd(bcc-A2)) in the Mn–Gd binary system is described with the substitutional solution model based on random mixing of the constituent atoms. Molar Gibbs energies of these solutions are expressed as follows:

$$\begin{aligned} G_{m}^{\phi} &= \sum_{i=Gd,Mn} x_{i}\,{}^{0}G_{i}^{\phi} + RT\sum_{i=Gd,Mn} x_{i}\ln x_{i} \\ &\quad + x_{Gd}x_{Mn}\sum_{j=0} {}^{(j)}L_{Gd,Mn}^{\phi}(x_{Gd} - x_{Mn})^{j} + {}^{mag}G_{m}^{\phi} \end{aligned} \tag{2}$$

$${}^{mag}G_{m}^{\phi} = RT\ln(\beta_{0} + 1)g(\tau) \tag{3}$$

where ${}^{0}G_{i}^{\phi}$ is the molar Gibbs energy of the element i in the respective reference state with the structure ϕ in a non-magnetic state, which is taken from the compilation by Dinsdale [30]. x_{i} means the mole fraction of the element i, T stands for the absolute temperature in Kelvin, and R is the gas constant. τ is the normalized temperature, which is expressed as $\tau = T/T_{C}^{\phi}$, and T_{C}^{ϕ} is the Curie temperature of phase ϕ. β_{0} is the number of Bohr magnetrons per mole. ${}^{ex}G_{m}^{\phi}$ is the excess Gibbs energy. ${}^{mag}G_{m}^{\phi}$ is the magnetic

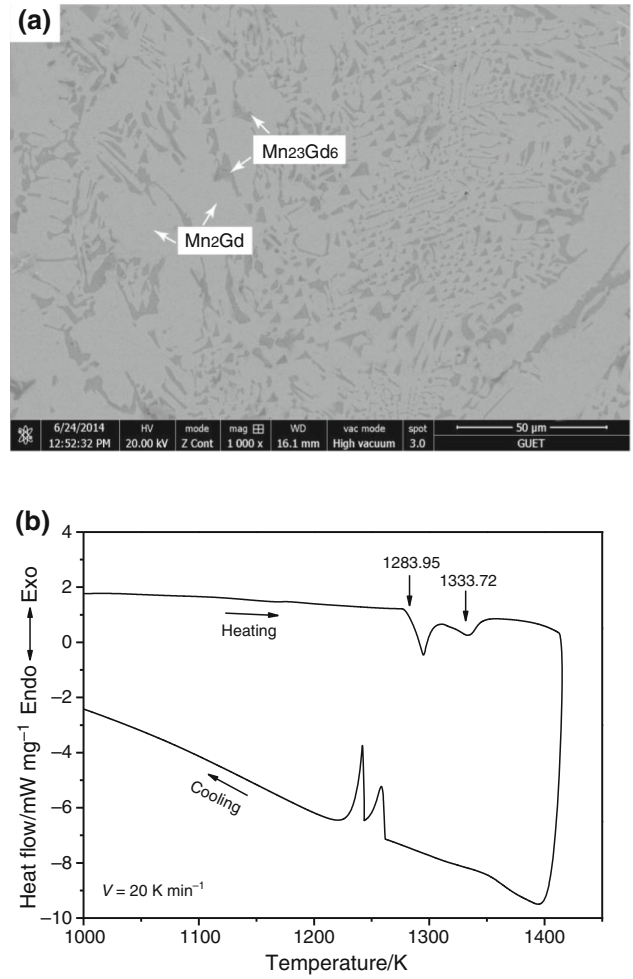

Fig. 1 SEM micrograph and thermal analysis curve of the as-cast $Mn_{70}Gd_{30}$ alloy

Table 1 Experimental results of the Mn–Gd alloys measured by thermal analysis

Sample composition/at.%	Phase transformation temperatures/K			
	Invariant reactions			Liquidus
Mn10Gd90	1092.84			–
Mn15Gd85	1094.34			–
Mn20Gd80	1093.50			–
Mn30Gd70	1093.33			–
Mn40Gd60	1096.17			–
Mn45Gd55	1089.50			–
Mn50Gd50	1091.41			1226.92
Mn55Gd45	1095.62			1229.95
Mn60Gd40	1099.27			1248.38
Mn65Gd35	1109.59	1271.41		–
Mn70Gd30 (as-cast)		1283.95		1333.72
Mn75Gd25		1278.53		1337.54
Mn80Gd20			1313.10	1340.37

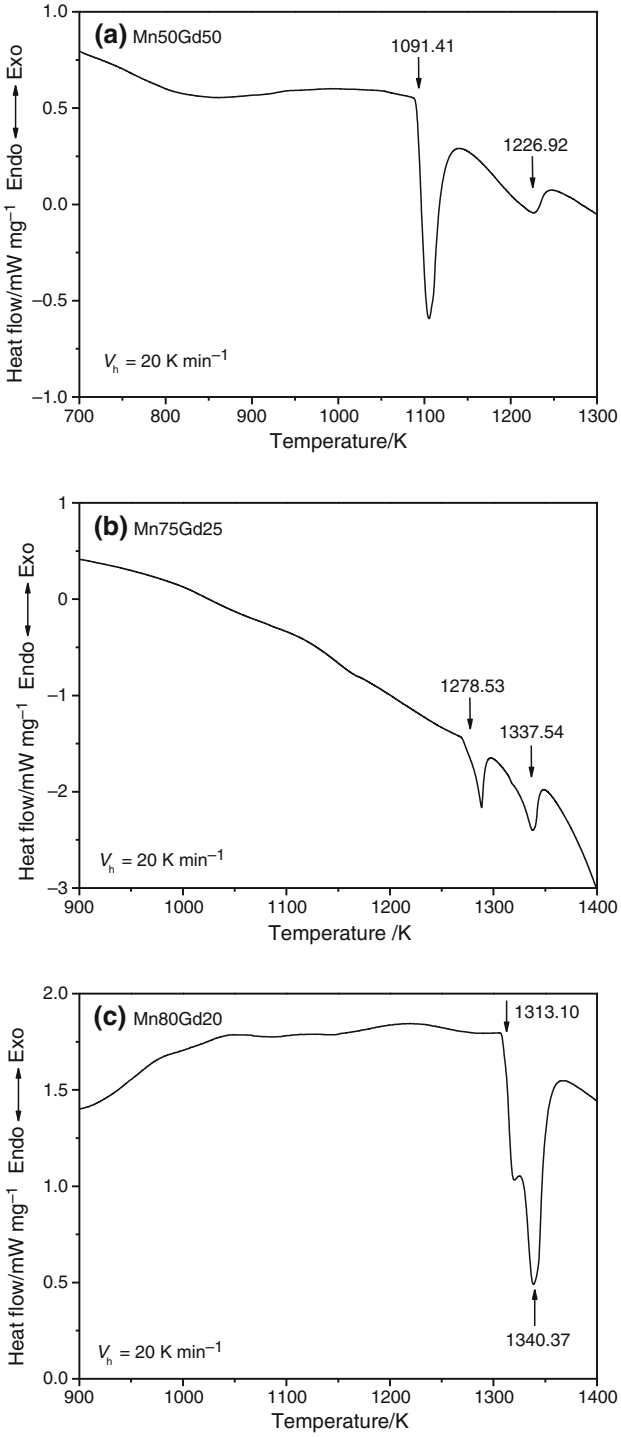

Fig. 2 Thermal analysis curves of the annealed Mn–Gd alloys at 20 K min^{-1} heating rate. **a** Mn50Gd50 alloy; **b** Mn75Gd25 alloy; **c** Mn80Gd20 alloy

contribution to the Gibbs energy of phase ϕ. ${}^{(j)}L^{\phi}_{\mathrm{Gd,Mn}}$ is the binary interaction parameter between two elements and can be expressed as:

$$ {}^{(j)}L^{\phi}_{\mathrm{Gd,Mn}} = A_{\mathrm{j}} + B_{\mathrm{j}}T \tag{4} $$

where A_{j} and B_{j} are the parameters to be optimized in this work.

Note that the lattice stabilities of Gd in γ-Mn(fcc) structure are not available in the SGTE database given by Dinsdale [30]. In the present work, the Gibbs energy of Gd in γ-Mn(fcc) structure (${}^{0}G^{\mathrm{fcc}}_{\mathrm{Gd}}$) is taken from the data reported by Liu et al. [31] as given:

$$ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Gd}} = 5000 + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} \tag{5} $$

Intermetallic compounds

The intermetallic compounds, $Mn_{12}Gd$, $Mn_{23}Gd_6$ and Mn_2Gd, are treated as the stoichiometric compound. Because the heat capacity of Mn_2Gd was determined experimentally by Marcos et al. [24], the molar Gibbs energy of Mn_2Gd can be expressed as:

$$ G^{\mathrm{Mn_2Gd}}_{\mathrm{m}} = \begin{cases} C_1 + D_2T + E_1T\ln T + F_1T^2 + {}^{\mathrm{mag}}G^{\mathrm{Mn_2Gd}}_{\mathrm{m}} & (T < 105) \\ 2\,{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} + C_2 + D_2T + E_2T\ln T + F_2T^2 + HT^{-1} + {}^{\mathrm{mag}}G^{\mathrm{Mn_2Gd}}_{\mathrm{m}} & (T > 105) \end{cases} \tag{6} $$

Due to the lack of experimental heat capacity data, using the Neumann–Kopp rule, the molar Gibbs energies of $Mn_{23}Gd_6$ and $Mn_{12}Gd$ are given as:

$$ G^{\mathrm{Mn_{23}Gd_6}}_{\mathrm{m}} = 23\,{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + 6\,{}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} + C_3 + D_3T + {}^{\mathrm{mag}}G^{\mathrm{Mn_{23}Gd_6}}_{\mathrm{m}} \tag{7} $$

$$ G^{\mathrm{Mn_{12}Gd}}_{\mathrm{m}} = 12\,{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} + C_4 + D_4T \tag{8} $$

The parameters (C_1, D_1, C_2, D_2, C_3, D_3, C_4, D_4, E_1, E_2, F_1, F_2 and H) mentioned above were to be optimized in this work. Note that the magnetic contribution to Gibbs energies of two intermetallic compounds, Mn_2Gd and $Mn_{23}Gd_6$ (${}^{\mathrm{mag}}G^{\mathrm{Mn_{23}Gd_6}}_{\mathrm{m}}$ and ${}^{\mathrm{mag}}G^{\mathrm{Mn_2Gd}}_{\mathrm{m}}$), is expressed by Eq. (3) due to their magnetic properties. In this work, the Curie temperatures (T_0) and Bohr magnetrons (β_0) of Mn_2Gd and $Mn_{23}Gd_6$ are 105 K and 4.8 μ_{B}, 461 K and 49 μ_{B} [24, 32], respectively.

Results and discussion

Experimental results

In the present work, thirteen Mn–Gd alloys were measured by means of thermal analysis. Different heating and cooling rates (10 and 20 K min^{-1}) were used in the present DTA measurements. The onset temperature of the thermal effect was selected in the heating curve as the temperature of the invariant reaction, while the peak temperature of the last thermal effect was used as the liquidus temperature.

Table 2 Thermodynamic parameters of the Mn–Gd binary system

Phase	Thermodynamic parameters[a]	References
Liquid	${}^{(0)}L^{\mathrm{Liq}}_{\mathrm{Gd,Mn}} = -7436.63 - 1.717T$	This work
	${}^{(1)}L^{\mathrm{Liq}}_{\mathrm{Gd,Mn}} = 3493.31 + 0.547T$	
	${}^{(2)}L^{\mathrm{Liq}}_{\mathrm{Gd,Mn}} = 1711.81 - 3.451T$	
cbcc_A12 (α-Mn)	${}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}}$ cited from SGTE database	[30]
cub_A13 (β-Mn)	${}^{0}G^{\mathrm{cub}}_{\mathrm{Mn}}$ cited from SGTE database	[30]
fcc_A1 (γ-Mn)	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Gd}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} + 5000$	[31]
	${}^{(0)}L^{\mathrm{fcc}}_{\mathrm{Gd,Mn}} = 34{,}000$	This work
bcc_A2 (δ-Mn, β-Gd)	${}^{(0)}L^{\mathrm{bcc}}_{\mathrm{Gd,Mn}} = 50{,}000$	This work
hcp_A3 (α-Gd)	${}^{(0)}L^{\mathrm{hcp}}_{\mathrm{Gd,Mn}} = 50{,}000$	This work
Mn_2Gd	$G^{\mathrm{Mn_2Gd}}_{\mathrm{m}} = \begin{cases} -26{,}867.49 - 11.649T + 0.9T\ln T - 0.29T^2 & (T<105) \\ 2\,{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - 6900 - 2.544T & \\ -2.5T\ln T + 0.007T^2 - 450{,}000T^{-1} & (T>105) \end{cases}$	This work
	$T^{\mathrm{Mn_2Gd}}_{\mathrm{C}} = 105,\ \beta^{\mathrm{Mn_2Gd}}_{0} = 4.8$	[24]
$Mn_{23}Gd_6$	$G^{Mn_{23}Gd_6}_{m} = 23\,{}^{0}G^{cbcc}_{Mn} + 6\,{}^{0}G^{hcp}_{Gd} - 75000 - 89.204T$	This work
	$T^{\mathrm{Mn_{23}Gd_6}}_{\mathrm{C}} = 461,\ \beta^{\mathrm{Mn_{23}Gd_6}}_{0} = 49$	[32]
$Mn_{12}Gd$	$G^{\mathrm{Mn_{12}Gd}}_{\mathrm{m}} = 12\,{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - 20{,}000 - 14.233T$	This work

[a] The Gibbs energies (J mol^{-1}) of pure Mn and Gd with liquid, bcc and hcp structures are given by Dinsdale [30]. The unit of temperature (T) is Kelvin (K)

The experimental results of thirteen Mn–Gd alloys determined by DTA measurements in this work are summarized in Table 1.

Figure 1 shows the SEM micrograph and thermal analysis curve of the as-cast $Mn_{70}Gd_{30}$ alloy. According to the EDS measurements, the as-cast alloy in Fig. 1a has the main phase (the gray phase, 67.21 at.% Mn, 32.79 at.% Gd) with the secondary phase (the dark phase, 79.83 at.% Mn, 20.17 at.% Gd), which are corresponding to Mn_2Gd and $Mn_{23}Gd_6$, respectively. The morphology of the as-cast alloy is not typically eutectic microstructure, which is deduced that the congruent melting of Mn_2Gd compound suggested by Antion [28] is impossible, while the formation of Mn_2Gd compound is formed through the peritectic reaction (L + $Mn_{23}Gd_6$ ↔ Mn_2Gd). Thermal analysis curve of the as-cast $Mn_{70}Gd_{30}$ alloy at 20 K min^{-1} heating/cooling rate is shown in Fig. 1b. There are two endothermic peaks and two exothermic peaks in the thermal analysis curves, which are responsible to be the peritectic reaction (L + $Mn_{23}Gd_6$ ↔ Mn_2Gd) at 1283.95 K and the liquidus temperature at 1333.72 K, respectively.

Figure 2 shows the representative thermal analysis curves of the annealed Mn–Gd alloys measured in the present work. In Fig. 2a, there are two endothermic peaks in the thermal analysis heating curve of the annealed $Mn_{50}Gd_{50}$ alloy, which are responsible to be the eutectic reaction (L ↔ Mn_2Gd + α-Gd) at 1091.41 K and the liquidus temperature at 1226.92 K, respectively. In Fig. 2b, two endothermic peaks were observed in the thermal analysis heating curve of the annealed $Mn_{75}Gd_{25}$ alloy, which is related to the peritectic reaction (L + $Mn_{23}Gd_6$ ↔ Mn_2Gd) at 1278.53 K and the liquidus temperature at 1337.54 K. According to two endothermic peaks in the thermal analysis heating curve of the annealed $Mn_{80}Gd_{20}$ alloy as shown in Fig. 2c, the temperature of the eutectic reaction (L ↔ $Mn_{23}Gd_6$ + $Mn_{12}Gd$) and the liquidus temperature were measured to be 1313.10 and 1340.37 K, respectively.

As shown in Table 1, in the present work, the temperatures of three invariant reactions in the Mn–Gd binary system were determined to be 1093.00 K (L ↔ Mn_2Gd + α-Gd), 1277.90 K (L + $Mn_{23}Gd_6$ ↔ Mn_2Gd) and 1313.10 K (L ↔ $Mn_{23}Gd_6$ + $Mn_{12}Gd$), respectively. On the other hand, it should be mentioned that the temperature of the peritectic reaction (L + γ-Mn ↔ $Mn_{12}Gd$) was not be determined in the present work. The liquidus temperatures of the annealed rich-Gd alloys (>55 at.%, Gd) were difficult to be detected due to the weak signal in thermal curves, resulting from the oxidation of rich-Gd liquid alloys. In addition, the liquidus temperature of annealed $Mn_{65}Gd_{35}$ alloy could not be measured because its liquidus temperature was very close to the temperature of the invariant reaction.

Table 3 Invariant reactions in the Mn–Gd binary system

Invariant reaction	Type	T/K	Composition/x^L_{Gd}	References
L ↔ α-Gd + Mn_2Gd	Eutectic	1103	64.7	[22]
		1101	66.2	[21]
		1086	–	[26]
		1093.0	–	This work (Exp)
		1092.2	66.9	This work (Cal)
L + $Mn_{23}Gd_6$ ↔ Mn_2Gd	Peritectic	1223	48	[22]
		1214	47.3	[21]
		1277.9	–	This work (Exp)
		1278.0	35.3	This work (Cal)
L ↔ $Mn_{23}Gd_6$ + $Mn_{12}Gd$	Eutectic	1325	20.5	[21]
		1311	–	[26]
		1313.1	–	This work (Exp)
		1313.8	13.3	This work (Cal)
L + β-Mn ↔ $Mn_{12}Gd$	Peritectic	1358	11	[22]
		1358	11.3	[21]
		1320	–	[26]
		1320.0	12.1	This work (Cal)
L + β-Gd ↔ α-Gd	Peritectic	1501	–	[26]
		1530.1	97.3	This work (Cal)
L + δ-Mn ↔ γ-Mn	Peritectic	1398	–	[22]
		1407	–	[21]
		1400	–	[26]
		1400.6	7.0	This work (Cal)
L + γ-Mn ↔ β-Mn	Peritectic	1360	–	[26]
		1357.6	9.6	This work (Cal)
L ↔ $Mn_{23}Gd_6$	Congruent	1329.0	–	[26]
		1340.0	–	This work (Cal)
$Mn_{12}Gd$ + β-Mn ↔ α-Mn	Peritectoid	980	–	[21]
		980.0	–	This work (Cal)

Thermodynamic calculation

Using the stabilities of the elements Mn and Gd compiled by Dinsdale [31], the model parameters of various phases in the Mn–Gd binary system were optimized using the PARROT module in the Thermo-Calc® software package developed by Sundman et al. [33] based on the experimental data reported in the literature and determined in the present work. This module works by minimizing the square sum of the differences between the experimental data and calculated values. In the optimization procedure, each set of the experimental data is given a certain weight according to the reliability and compatibility of the experimental data. It should be claimed that the weights of much more reliable experimental data are larger than those of less ones during the optimization. The weight can be changed systematically during the optimization until most of the experimental data are accounted for within the claimed uncertainty limits. Thermodynamic parameters of all condensed phases in the Mn–Gd binary system used and obtained finally in the present work are summarized in Table 2. The calculated temperatures and compositions of all invariant reactions in this binary system are given in Table 3.

Figure 3 shows the calculated phase diagram of the Mn–Gd binary system in this work. The phase diagram calculated in this work is compared with the calculated results by Gröbner et al. [21] as given in Fig. 4. Figure 5 shows the comparison of the calculated phase diagram with the experimental data determined in the present work and reported by Kirchmayr and Lugscheider [22] and Antion [25]. The calculated liquidus in the rich-Mn part (>55 at.% Mn) is in good agreement with the experimental data determined in the present work and reported by Kirchmayr

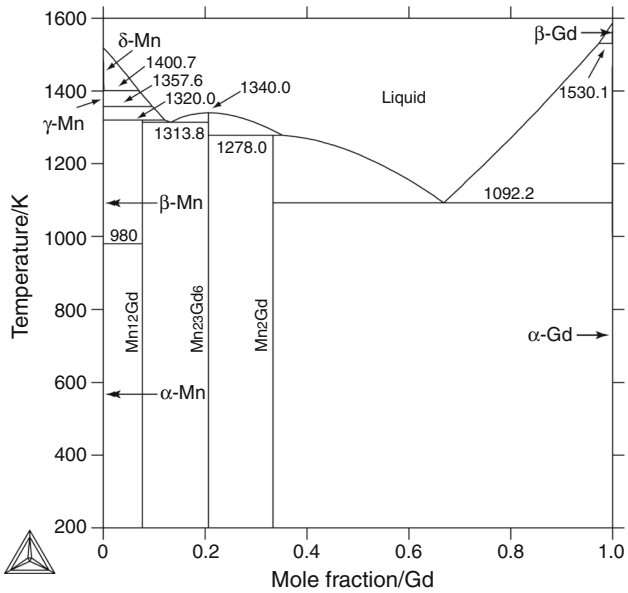

Fig. 3 Calculated phase diagram of the Mn–Gd binary system in this work

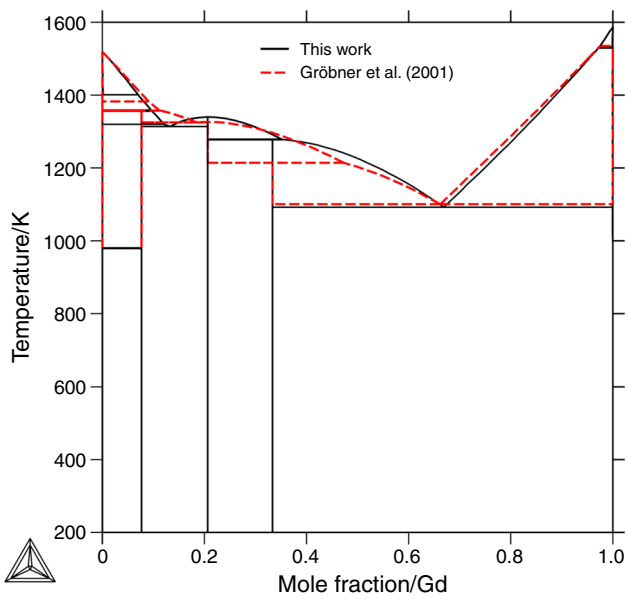

Fig. 4 Comparison of the calculated phase diagram of the Mn–Gd binary system in this work with the calculated results [21]

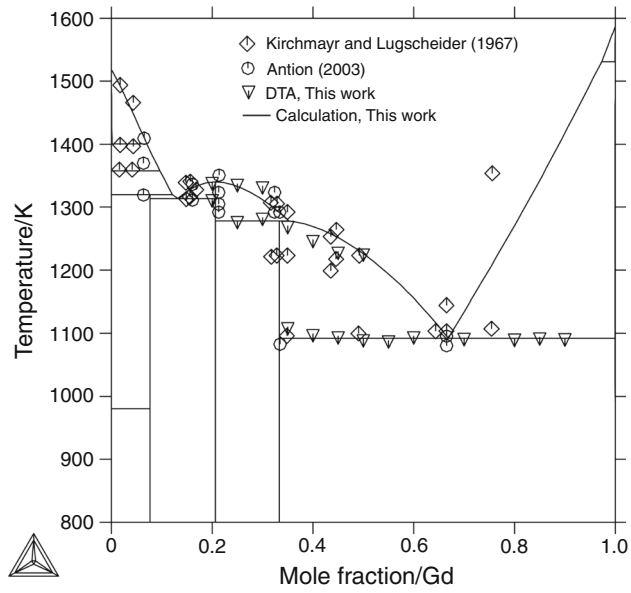

Fig. 5 Calculated phase diagram of the Mn–Gd system with the experimental data determined in the present work and reported in Refs. [22, 25]

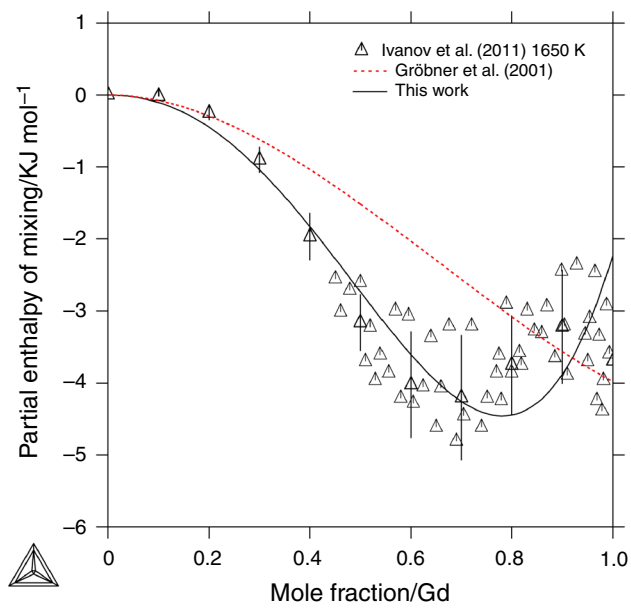

Fig. 6 Calculated partial enthalpy of Mn in liquid alloys at 1650 K with the experimental data [23] (Reference state: liquid Mn)

and Lugscheider [22] and Antion [25], while there is obvious difference between the calculated liquidus in the rich-Gd part and the experimental data [22]. It is still reasonable because the liquidus determined by Kirchmayr and Lugscheider [22] in the rich-Gd part is not available, which was pointed out by Okamoto [28]. The calculated temperatures and compositions of the invariant reactions in this binary system are compared with the experimental data and the previously calculated results [22, 26] as given in Table 3. As can be seen, the calculated temperatures of the invariant reactions agree well with the experimental data determined in the present work within experimental error (about 1 K). However, the calculated temperature (1092.20 K) of eutectic reaction ($L \leftrightarrow Mn_{23}Gd_6 + Mn_2Gd$) is lower than the value (1103.0 K) determined by Kirchmayr and Lugscheider [22] and is higher than the data (1083.0 K) calculated by Kim and Jung [26]. The calculated temperature (1278.0 K) of the peritectic reaction ($L + Mn_{23}Gd_6 \leftrightarrow Mn_2Gd$) is much higher than the data

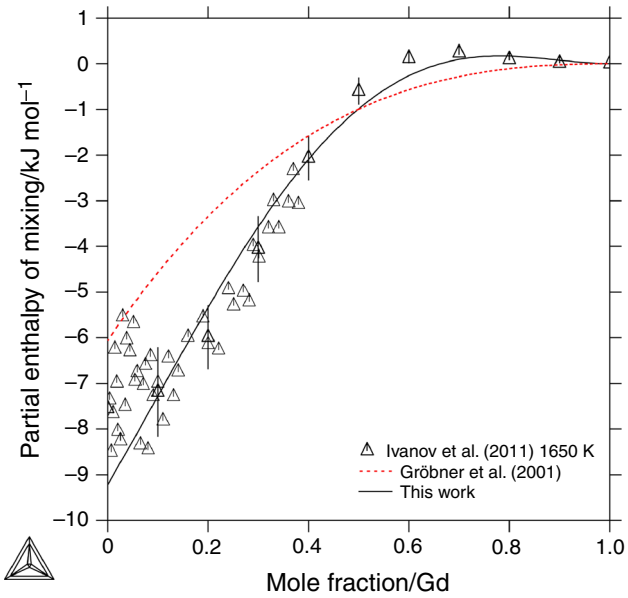

Fig. 7 Calculated partial enthalpy of Gd in liquid alloys at 1650 K with the experimental data [23] (Reference state: liquid Gd)

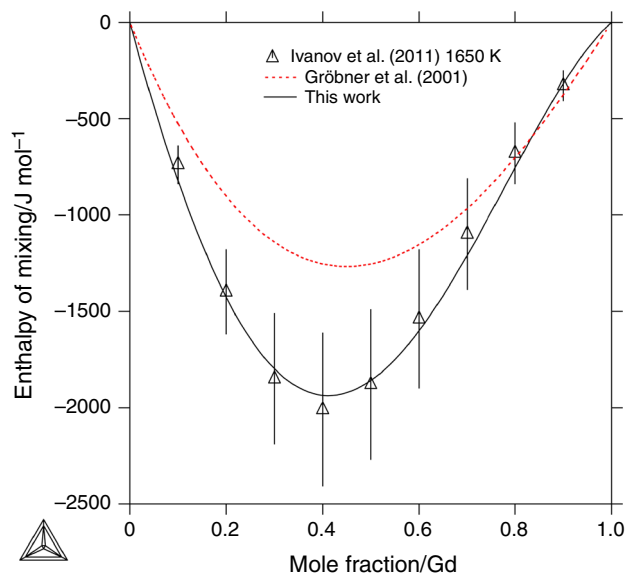

Fig. 8 Calculated enthalpy of mixing of liquid Mn–Gd alloys at 1650 K with the experimental data [23] (Reference states: liquid Mn and liquid Gd)

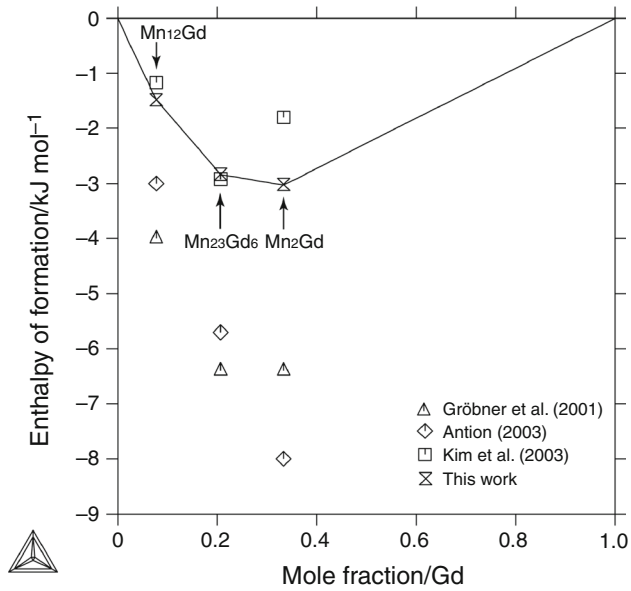

Fig. 9 Calculated enthalpies of formation of intermetallic compounds at 298 K with the assessed results [21, 25, 26] (Reference states: α-Mn and α-Gd)

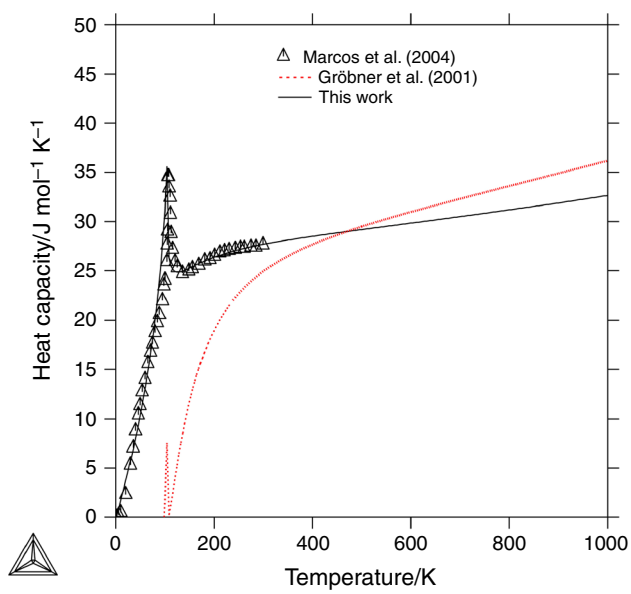

Fig. 10 Calculated heat capacity of Mn_2Gd with the experimental data [24]

(1223.0 K) measured by Kirchmayr and Lugscheider [22] and the value (1214 K) calculated by Gröbner et al. [21]. The calculated temperature (1313.8 K) of the eutectic reaction ($L \leftrightarrow Mn_{23}Gd_6 + Mn_2Gd$) agrees well with the value (1311.0 K) calculated by Kim and Jung [26] and is lower than the data (1325.0 K) calculated by Gröbner et al. [21].

Figures 6 and 7 show the calculated partial enthalpies of mixing of Gd and Mn in liquid Mn–Gd alloys at 1650 K with the experimental data measured by Ivanov et al. [23] and the calculated results [21]. The calculated enthalpies of mixing of liquid alloys at 1650 K with the experimental data [23] and the calculated results [21] are given in Fig. 8. It can be seen that the calculated results in this work are satisfactorily consistent with the experimental data within the estimated error, while the calculated results by Gröbner et al. [21] are not obviously consistent.

Figure 9 is the calculated enthalpies of formation of the intermetallic compounds in the Mn–Gd binary system at 298 K with the assessed results by Gröbner et al. [21],

Antion [25] and Kim and Jung [26]. In this work, the enthalpies of formation of the intermetallic compounds, $Mn_{12}Gd$, $Mn_{23}Gd_6$ and Mn_2Gd, were calculated to be −1.48, −2.83 and −3.01 kJ mol^{-1}, respectively, which are much more positive than the data assessed previously by Gröbner et al. [21] and Antion [25]. Using modified Miedema model, the enthalpies of formation of $Mn_{12}Tb$, $Mn_{23}Tb_6$ and $MnTb_2$ at 298 K were calculated by Kim and Pailwal [34] to be −1.05, −2.51 and −3.19 kJ mol^{-1}. Boer et al. [35] indicated that the enthalpies of formation of the intermetallic compounds in the Mn–RE binary systems are between −1000 and −2000 J mol^{-1}. Hence, considering the similar chemical/physical properties of rare earth elements, the calculated results in the present work are reasonable. It should be noticed that the further experimental measurements of the enthalpies of formation of the intermetallic compounds in the Mn–Gd binary system are still necessary.

Figure 10 shows the calculated heat capacity of Mn_2Gd with the experimental data by Marcos et al. [24] and the calculated results by Gröbner et al. [21]. As can be seen, the calculated results in this work are in good agreement with the experimental data measured by Marcos et al. [24], while there is obvious difference between the calculated results by Gröbner et al. [21] and the experimental data [24].

Conclusions

The phase diagram of the Mn–Gd binary system was measured through thirteen alloys prepared by arc-melting method with differential thermal analysis. The temperatures of the invariant reactions and liquidus in the Mn–Gd binary system were determined. In particular, the type of the invariant reaction, $L + Mn_{23}Gd_6 \leftrightarrow Mn_2Gd$, was confirmed using one key as-cast $Mn_{70}Gd_{30}$ alloy with SEM/EDS and DTA analysis. The Mn–Gd binary system was thermodynamically optimized on the basis of experimental thermodynamic data and phase equilibria data determined in the present work and published literature. A set of self-consistent thermodynamic parameters has been obtained. The calculated phase diagram and thermodynamic properties agree well with the experimental data.

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